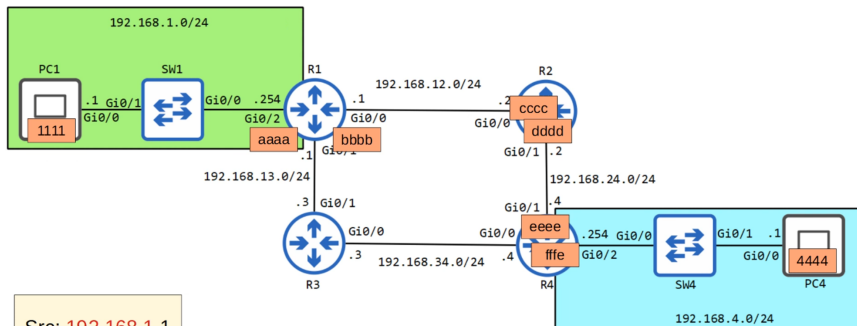


Day 12 - The life of a packet

Saturday, August 22, 2026 2:50 PM

12. LIFE OF A PACKET

Network Topology



EACH Network device's interfaces have a UNIQUE MAC Addresses.

In a TCP HEADER

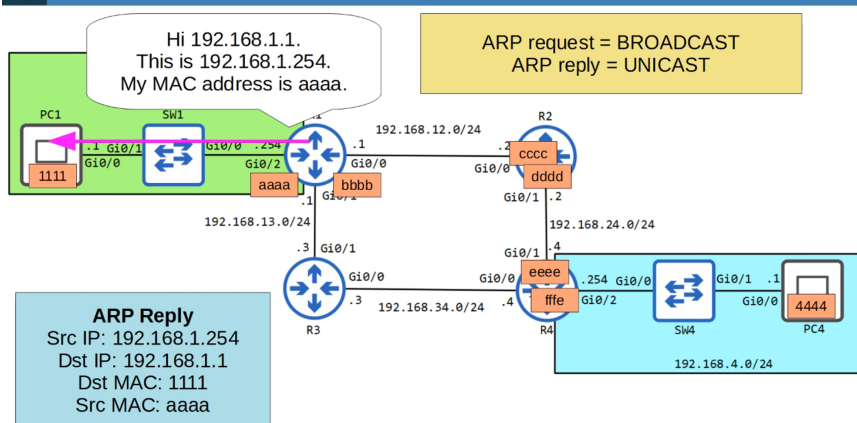
SOURCE IP ADDRESS comes before DESTINATION IP ADDRESS

while...

in an ETHERNET HEADER

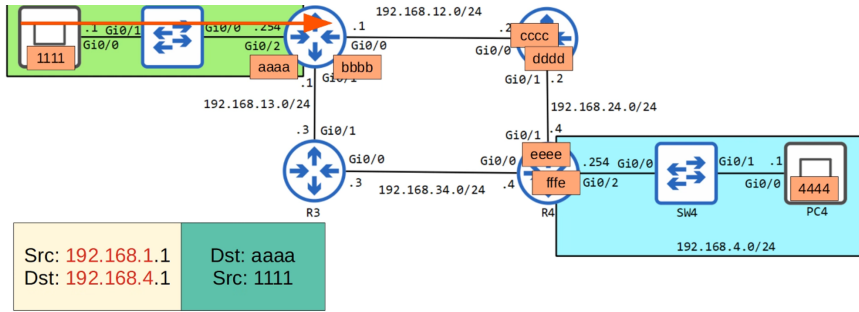
DESTINATION MAC ADDRESS comes before SOURCE MAC ADDRESS

ARP (Address Resolution Protocol)

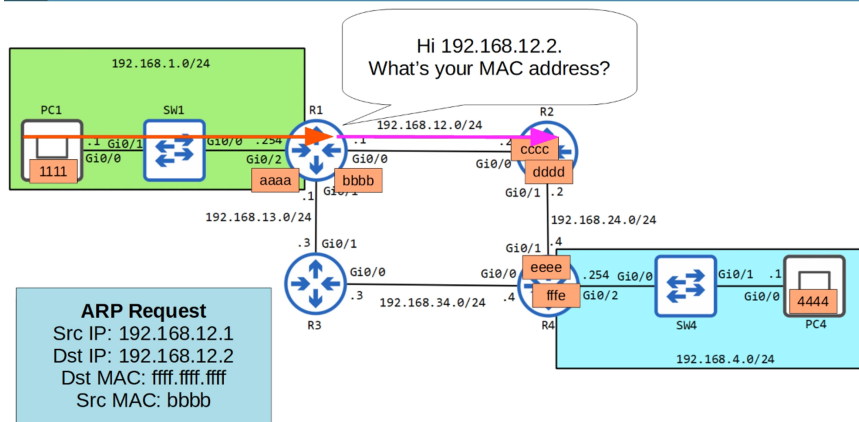


PC1 → R1

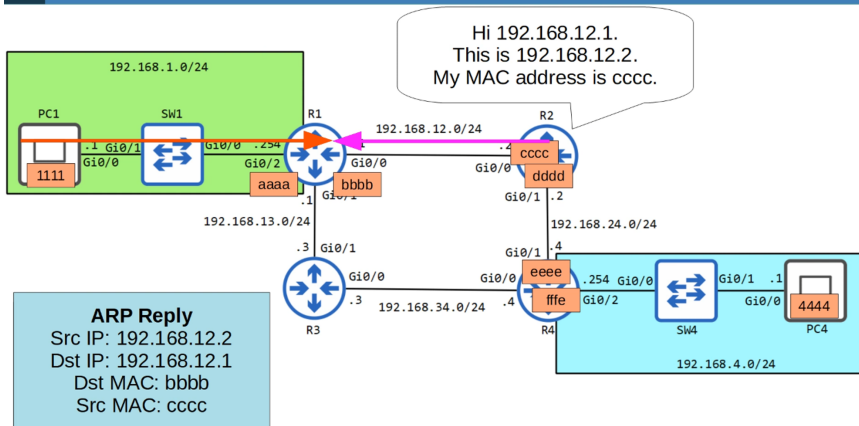




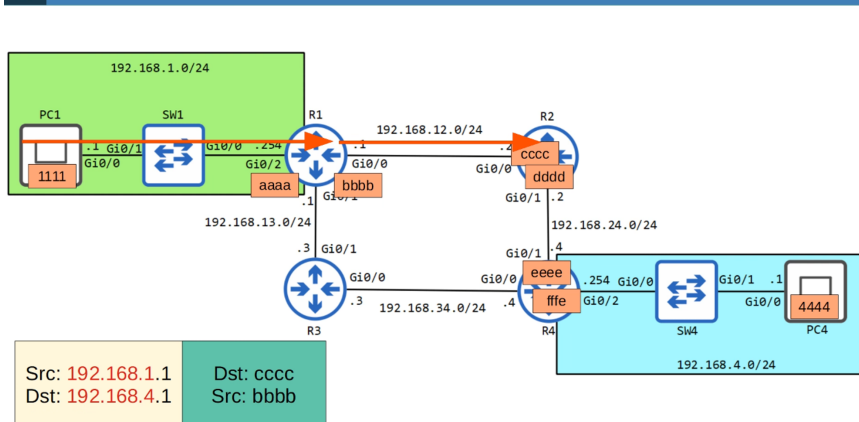
ARP



ARP

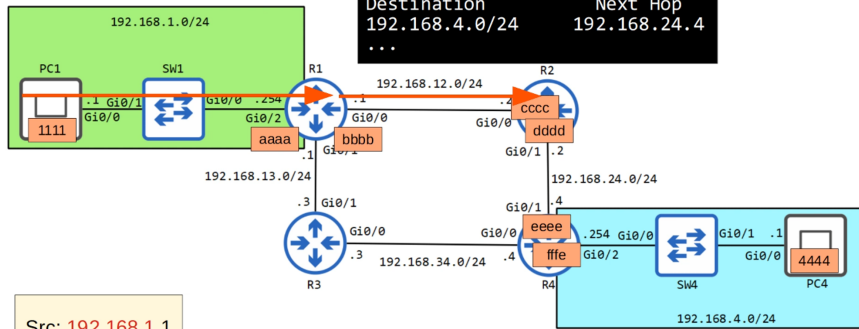


R1 → R2



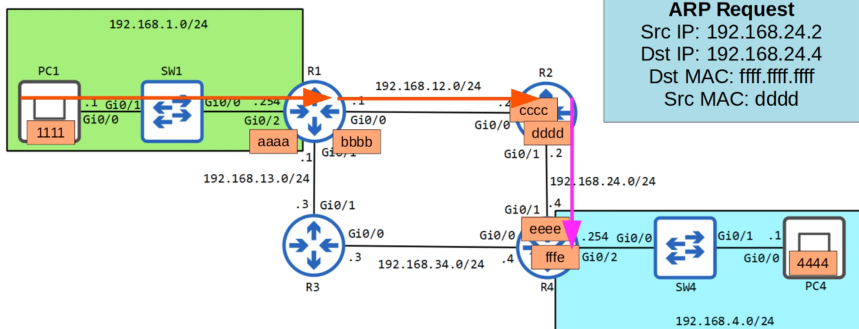
R2 → R4

R2 Routing Table	
Destination	Next Hop
192.168.4.0/24	192.168.24.4
...	



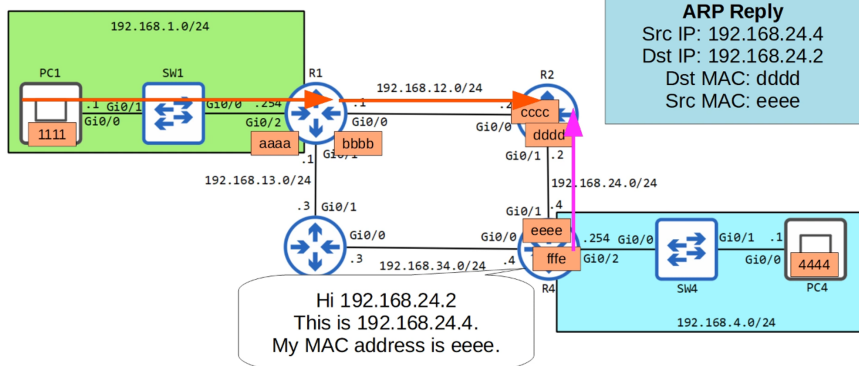
Src: 192.168.1.1
Dst: 192.168.4.1

ARP



ARP Request
Src IP: 192.168.24.2
Dst IP: 192.168.24.4
Dst MAC: ffff.ffff
Src MAC: dddd

ARP



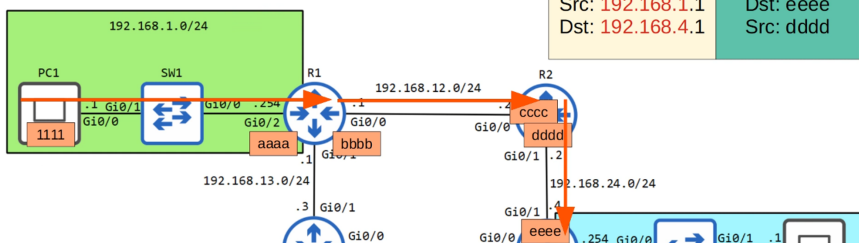
ARP Reply
Src IP: 192.168.24.4
Dst IP: 192.168.24.2
Dst MAC: dddd
Src MAC: eeee

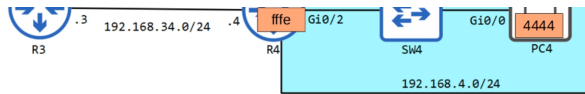
Hi 192.168.24.2
This is 192.168.24.4.
My MAC address is eeee.

R2 → R4

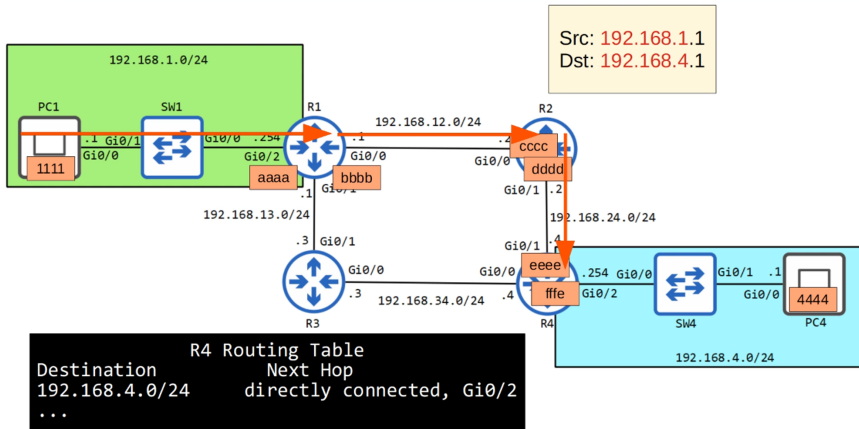
Src: 192.168.1.1
Dst: 192.168.4.1

Dst: eeee
Src: dddd

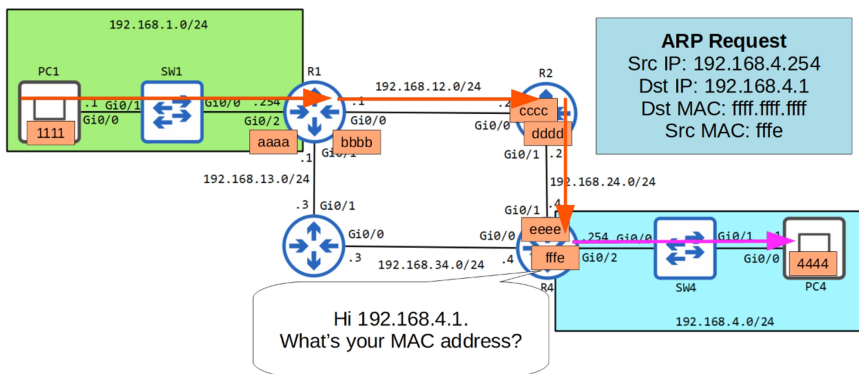




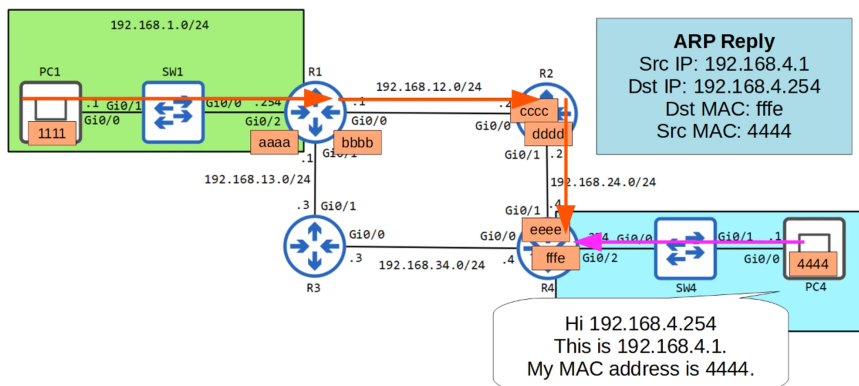
R2 → R4



ARP

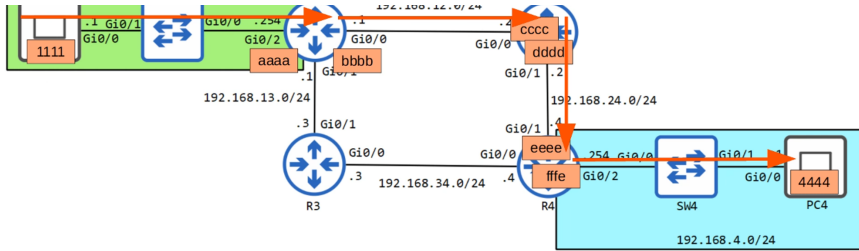


ARP

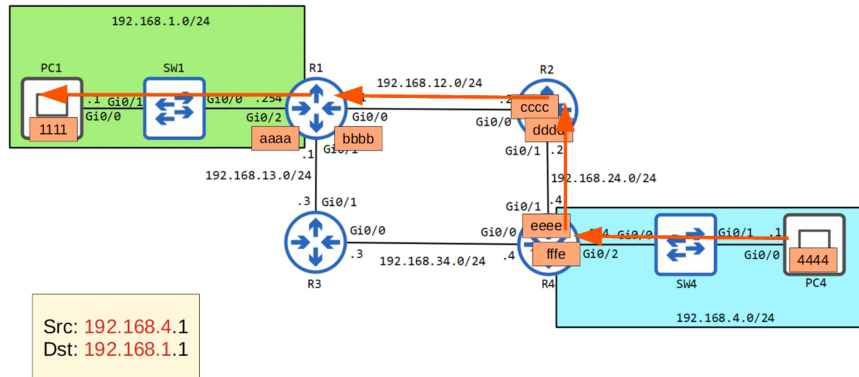


R4 → PC4





R4 → PC4



When a HOST sends a packet to another HOST, the SOURCE or DESTINATION IP doesn't change - even though ROUTERS may change the ETHERNET HEADER (SRC/DEST MAC ADDRESSES).